

Cai Zhou

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Second-year PhD at MIT EECS. Advisors: Tommi Jaakkola, Stephen Bates.

Education Background

- **Massachusetts Institute of Technology** **Cambridge, MA**
PhD in EECS, CSAIL & LIDS *2024-2029*
- **Tsinghua University** **Beijing**
Bachelor in Department of Automation, Class of General Artificial Intelligence *2020-2024*
GPA: 3.93/4.0, Ranking: Top 2%
- **Tsinghua University** **Beijing**
Bachelor in Statistics, Secondary Degree *2021-2024*

Research Interest

My research interest lies broadly in **machine learning** and **generative models**. My recent research focuses on (i) generative models such as **diffusion models**, with applications to vision, language, and biomolecule; (ii) **language models pretraining**, including architectures and scaling; (iii) **LLM reasoning** and test-time scaling paradigms, including diffusion language models and latent reasoning.

Internship Experience

- **Apple Machine Learning Research (Apple MLR)** **Cupertino, CA**
Mentored by Yizhe Zhang and Ruixiang Zhang, Part-time *2026.1 - Present*
- **Microsoft Research (MSR)** **Cambridge, MA**
Mentored by Dinghuai Zhang and Lester Mackey *2025.5 - 2025.8*

Selected Publications

Full list of publications available on my [homepage](#).

Preprints.....

- Online Reasoning Calibration: Test-Time Training Enables Generalizable Conformal LLM Reasoning [PDF]
Cai Zhou*, Zekai Wang*, Menghua Wu, ..., Chenyu Wang, Ashia Wilson, Tommi Jaakkola, Stephen Bates
TL;DR: We propose Online Reasoning Calibration (ORCA), integrating the Test-Time Training (TTT) framework into conformal prediction, enabling generalizable and risk-controlled early stopping. Significantly improves LLM reasoning efficiency by over 60% in OOD setting.
- Rethinking Diffusion Models with Symmetries through Canonicalization with Applications to Molecular Graph Generation [PDF]
Cai Zhou*, Zijie Chen*, ..., Kaiyi Jiang, Pan Li, Rose Yu, Muhan Zhang, Stephen Bates, Tommi Jaakkola
TL;DR: Theoretically show that canonicalization reduces conditional variance in diffusion and flow matching with symmetries, accelerating model training. Improves molecular graph generation quality with no overheads.
- What Affects the Effective Depth of Large Language Models? [PDF]
Yi Hu, **Cai Zhou**, Muhan Zhang
TL;DR: We systematically study how effective depth varies with model scale, training type, and task difficulty. We found that the effective depth ratio remains stable across base and reasoning models with long-CoT in various tasks, suggesting underuse of layer depth in current LLMs.
- CLAP: Contrastive Latent Action Pretraining for Learning Vision-Language-Action Models from Human Videos
Chubin Zhang, Jianan Wang, Zifeng Gao, Yue Su, Tianru Dai, **Cai Zhou**, Jiwen Lu, Yansong Tang [PDF]
TL;DR: We propose CLAP, which aligns VLA executable latent action space with visual latent space of human videos in VLA pretraining, enabling the effective transfer of skills.

Conference and Journal Papers.....

- Coevolutionary Continuous Discrete Diffusion: Make Your Diffusion Language Model a Latent Reasoner
Cai Zhou, Chenxiao Yang, Yi Hu, Chenyu Wang, Chubin Zhang, Muhan Zhang, Lester Mackey, Tommi Jaakkola, Stephen Bates, Dinghuai Zhang
Forty-third International Conference on Machine Learning (ICML), 2026. [PDF]
TL;DR: Analyze the theoretical expressivity and practical trainability of continuous diffusion, discrete diffusion and looped transformers. Propose CCDD, a diffusion language model combining the advantages of both continuous and discrete diffusion, which is expressive and easy to train, outperforming AR, DLM and LT in reasoning tasks.
- On Powerful Ways to Generate: Autoregression, Diffusion, and Beyond
Chenxiao Yang, **Cai Zhou**, David Wipf, Zhiyuan Li
Fourteenth International Conference on Learning Representations (ICLR), 2026. [PDF]
TL;DR: Theoretically analyze the expressiveness of autoregression model, diffusion language model, and more generalized any-process generation.
- SPG: Sandwiched Policy Gradient for Masked Diffusion Language Models
Chenyu Wang, Paria Rashidinejad, DiJia Su, Song Jiang, Sid Wang, Siyan Zhao, **Cai Zhou**, Shannon Zejiang Shen, Feiyu Chen, Tommi Jaakkola, Yuandong Tian, Bo Liu
Fourteenth International Conference on Learning Representations (ICLR), 2026. [PDF]
TL;DR: We propose the Sandwiched Policy Gradient (SPG) that leverages both an upper and a lower bound of the true log-likelihood of DLM in RL, outperforming previous RL baselines.
- Next Semantic Scale Prediction via Hierarchical Diffusion Language Models
Cai Zhou*, Chenyu Wang*, Dinghuai Zhang*, Shnagyan Tong, Yifei Wang, Stephen Bates, Tommi Jaakkola
Thirty-ninth Conference on Neural Information Processing Systems (NeurIPS), 2025. [PDF]
TL;DR: Propose HDLM, which introduces intermediate hierarchies for masked discrete diffusion model, and progressively predicts the next semantic scale tokens in language generation.
- Learning Diffusion Models with Flexible Representation Guidance
Chenyu Wang*, **Cai Zhou***, Sharut Gupta, Zongyu Lin, Stefanie Jegelka, Stephen Bates, Tommi Jaakkola
Thirty-ninth Conference on Neural Information Processing Systems (NeurIPS), 2025. [PDF]
TL;DR: Systematically analyze representation-enhanced diffusion model training. Propose REED, which consists of multimodal representation alignment, improved curriculum and domain-specific designs for generalized diffusion model representation guidance.
- Thought Calibration: Efficient and Confident Test-Time Scaling
Menghua Wu, **Cai Zhou**, Stephen Bates, Tommi Jaakkola
The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP main) **Oral** [PDF]
TL;DR: Proposal thought calibration, an inference-time calibration strategy for efficient test-time scaling via early stopping based on conformalized confidence score.
- Geometric Representation Condition Improves Equivariant Molecule Generation
Zian Li*, **Cai Zhou***, Xiyuan Wang, Xingang Peng, Muhan Zhang.
Forty-second International Conference on Machine Learning (ICML), 2025 **Spotlight (top 2.6%)** [PDF]
TL;DR: Propose GeoRCG, a general framework to enhance the performance of molecular generative models by integrating geometric representation conditions. Provide theoretical guarantees and achieve SOTA performance on several molecule generation tasks.
- ChemVLM: Exploring the Power of Multimodal Large Language Models in Chemistry Area
..., **Cai Zhou**, ...
Thirty-Ninth AAAI Conference on Artificial Intelligence (AAAI), 2025 [PDF]
TL;DR: Introduce ChemVLM, an open-source chemical multimodal large language model specifically designed for chemical applications, with ability to understand both textual and visual chemical information, including molecular structures, reactions, and chemistry examination questions.
- Unifying Generation and Prediction on Graphs with Latent Graph Diffusion
Cai Zhou, Xiyuan Wang, Muhan Zhang.
Thirty-eighth Conference on Neural Information Processing Systems (NeurIPS), 2024 [PDF]
TL;DR: Propose Latent Graph Diffusion, a generative framework for both generation and prediction tasks utilizing latent graph diffusion. Derive theoretical guarantees for solving regression and classification tasks with diffusion models. Strong performance in both generation and prediction tasks for molecules and general graphs.
- On the Theoretical Expressive Power and Design Spaces of High Order Graph Transformers

Cai Zhou, Rose Yu, Yusu Wang

Twenty-seventh International Conference on Artificial Intelligence and Statistics (AISTATS), 2024 [PDF]

TL;DR: Theoretically analyze the expressive power and approximation power of high order graph transformers; propose scalable and powerful high order graph transformers and simplicial transformers.

- Facilitating Graph Neural Networks with Random Walk on Simplicial Complexes

Cai Zhou, Xiyuan Wang, Muhan Zhang.

Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS), 2023 [PDF]

TL;DR: Design positional and structural encodings for GNNs based on random walk on simplicial complexes, specifically EdgeRWSE and Hodge1Lap for 1-simplices (edge-level) that can provably improve expressive power and practical performance.

- From Relational Pooling to Subgraph GNNs: A Universal Framework for More Expressive Graph Neural Networks

Cai Zhou*, Xiyuan Wang*, Muhan Zhang.

Fortieth International Conference on Machine Learning (ICML), 2023 [PDF]

TL;DR: Propose k, l -WL algorithm (running k -WL on a graph with l labels) and theoretically establish a strict expressive power hierarchy; Incorporate a wide range of GNNs including relational pooling and subgraph GNNs.

Crucial Honors and Awards

- Outstanding Graduate by Beijing Ministry of Education and by Tsinghua University
- Outstanding Undergraduate Thesis by Beijing Ministry of Education and by Tsinghua University, 2024
- National Scholarship (Highest honor for undergraduates in China), 2023
- Comprehensive Excellence Award of Tsinghua University (Highest honor in THU and the Dept. Automation), 2023 & 2021
- Research and Innovation Excellence Award of Tsinghua University, 2022
- Academic Excellence Award of Tsinghua University, 2022
- First prize of National Physics Competition for Undergraduates, 2021

Technical and General Skills

- **Mathematics:** Calculus, Algebra, Discrete and Combinatorial Mathematics, Differential Geometry, Algebraic Topology, Complex Analysis, Fourier Analysis, Functional Analysis, Operation Research, etc.
- **Statistics:** Probability Theory, Statistical Inference, Stochastic Process, Multivariate Statistics and Regression, Computational Statistics, Time Series Analysis, Causal Inference, Bayesian Statistics, Biostatistics, etc.
- **Programming Skills:** Proficient in Python, Pytorch, PyG, DGL, C, C++, R, Linux, Latex, Markdown, Git, etc. Familiar with Tensorflow, Keras, C#, HTML, MATLAB, etc.
- **General Skills:** Good writing and presentation skills; Outstanding leadership and cooperation skills.